Fair Early Warning in Online Education with Dynamic Bayesian Networks

ICT 5143: AFGML Capstone | Problem 3

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Research question and decision context

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- Question: can a temporally regularized alert rule reduce disparities in early support while keeping detection and a fixed capacity?
- Hypothesis: at 20% capacity, lower cumulative coverage gap than per-checkpoint equalisation, with detection falling by at most 0.02. Refuted if the gap is not lower or detection falls by more than 0.02.

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- Open issue: equalising each checkpoint separately ignores the cumulative alert history and timing of support.
- Contribution: joint tuning of group offsets across checkpoints, a re-alert discount, and smoothness penalties, under a fixed capacity.
- Evidence: comparison with the baseline, ablations that remove each component, and a transfer presentation.

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- Prediction-time information: enrolment data, weekly activity up to the checkpoint, assessments due and submitted by then. No future events, no assessment scores.

Graphical model and inference

$$C \to Z_1 \to Z_2 \to \cdots \to Z_8 \to Y$$

each Z_t (latent engagement) emits V_t (weekly activity) and Q_t (assessment behind)

$$p(C, Z, V, Q, Y) = \pi(Z_1 \mid C) \prod_t A(Z_t \mid Z_{t-1}) \prod_t B_V(V_t \mid Z_t)\, B_Q(Q_t \mid Z_t)\, \theta(Y \mid Z_T, C)$$

- Query and algorithm: $P(Y = 1 \mid C, V_{1:t}, Q_{1:t})$ by forward filtering; EM for parameters; three latent states.

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- Assumptions: V_t and Q_t independent given Z_t; Z_t is a statistical summary, not ability; no causal claim.
- Validation: brute force, three elimination orders and clique tree agree with the filter to 3.1×10^{-15} on 20 test students at four checkpoints. Tolerance 10^{-10}, passed.

Fairness criterion and proposed method

- Primary fairness criterion: cumulative coverage gap, the max−min of $P(\geq 1 \text{ alert} \mid Y = 1)$ over supported groups; checkpoint ΔEOpp as a secondary criterion. Rationale: the alert history is what students experience.

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- Diagnostics: false-alert exposure, PPV, time to first alert, ΔEOdds, calibration by group.

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- Mechanism tests: four ablations (no transitions, no smoothing, no latent state, no re-alert discount) and robustness to schedules, two states, 30% activity dropped, and alternative outcomes.

Main result and uncertainty

Method (test, 20%, five-split mean)	Detection (higher better)	Coverage gap (lower better)	Lo
Baseline (`indep_eopp`)	0.567 (0.017)	0.114 (0.096)	0.
Proposed (`temporal`)	0.683 (0.076)	0.095 (0.044)	0.
Top-k (no fairness adjustment)	0.568 (0.026)	0.088 (0.067)	0.

Paired effect vs baseline, test, mean over five splits: detection $+0.116$ (non-inferior in 5 of 5 splits); coverage gap -0.019 (significant in 0 of 5); false-alert exposure $+0.134$ (margin 0.05 held in 1 of 5).

Answer to the question: The temporal rule raises detection in every split and lowers the mean coverage gap versus per-checkpoint equalisation, but the gap reduction is not reliable on test, exposure nearly doubles, and the rule does not beat top-k on fairness.

Results

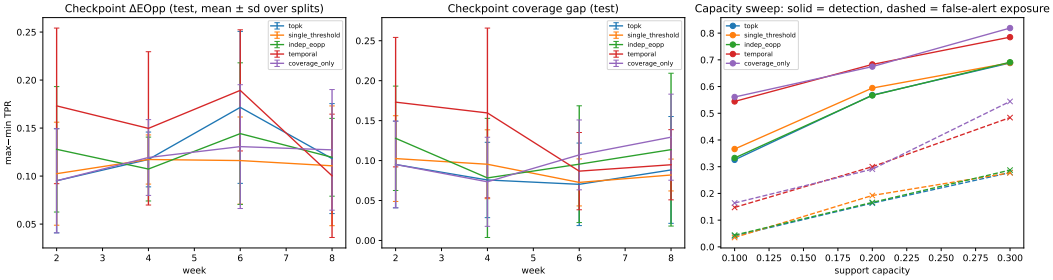


Figure: Result

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- Interpretation: split-to-split spread is large (gap sd 0.096 for the baseline). The five model seeds give virtually identical test results (seed 0 differs slightly on other policies), so seed stability is supported but not yet fully verified.

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- Held-out transfer and low-activity or missing-assessment slices: done.
- Calibration, log loss and Brier: done. Five model seeds: run, nearly identical. Tuning budgets: 200 vs 315, unequal. Group-input alert change: 10–30% of alerts (test, 20% capacity).

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- References: Hardt, Price and Srebro (2016); Kuzilek, Hlosta and Zdrahal (2017); Koller and Friedman (2009).

Backup: statistical validation and exact inference

Set	Baseline	Detection	Gap	Sig. splits
Test	`indep_eopp`	$+0.116$	-0.019	5 / 0
Test	`topk`	$+0.115$	$+0.006$	5 / 0
Transfer	`indep_eopp`	$+0.166$	-0.114	5 / 1
Transfer	`topk`	$+0.163$	$+0.036$	5 / 0

Temporal minus baseline, mean over five splits. Sig.: detection non-inferiority (H2) / gap significance (H1).

Simulator output, test set, all capacities (first 30 rows):

	policy	capacity	week	eligible	alerted	cum_detection	cum_coverage_gap	checkpoint_dEOpp
0	topk	0.1	2	945	94	0.183	0.096	0.096
1	topk	0.1	4	931	93	0.271	0.106	0.132
2	topk	0.1	6	915	92	0.294	0.101	0.134
3	topk	0.1	8	897	90	0.315	0.116	0.139
4	indep_eopp	0.1	2	945	94	0.181	0.053	0.053
5	indep_eopp	0.1	4	931	93	0.271	0.367	0.373
6	indep_eopp	0.1	6	915	92	0.308	0.352	0.165
7	indep_eopp	0.1	8	897	90	0.326	0.328	0.089
8	temporal	0.1	2	945	94	0.183	0.080	0.080
9	temporal	0.1	4	931	93	0.343	0.099	0.034
10	temporal	0.1	6	915	92	0.500	0.113	0.014
11	temporal	0.1	8	897	90	0.609	0.109	0.148
12	topk	0.2	2	945	189	0.333	0.182	0.182
13	topk	0.2	4	931	186	0.442	0.112	0.109
14	topk	0.2	6	915	183	0.519	0.058	0.265
15	topk	0.2	8	897	179	0.593	0.084	0.060
16	indep_eopp	0.2	2	945	189	0.333	0.192	0.192
17	indep_eopp	0.2	4	931	186	0.444	0.209	0.113
18	indep_eopp	0.2	6	915	183	0.512	0.215	0.081
19	indep_eopp	0.2	8	897	179	0.583	0.263	0.136
20	temporal	0.2	2	945	189	0.333	0.192	0.192
21	temporal	0.2	4	931	186	0.514	0.307	0.039
22	temporal	0.2	6	915	183	0.662	0.108	0.107
23	temporal	0.2	8	897	179	0.778	0.090	0.011
24	topk	0.3	2	945	284	0.470	0.167	0.167
25	topk	0.3	4	931	279	0.562	0.144	0.096
26	topk	0.3	6	915	274	0.613	0.138	0.102
27	topk	0.3	8	897	269	0.694	0.181	0.029

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- Subgroup denominators (test, positives/negatives): F|N 330/426; F|Y 41/35; M|N 54/45; M|Y 7/7.

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- Subgroup denominators (test, positives/negatives): F|N 330/426; F|Y 41/35; M|N 54/45; M|Y 7/7.
- Bootstrap: student-level, 1,000 replicates; Holm across H1 to H4. Non-inferiority margins: detection 0.02; false-alert exposure 0.05.

Backup: demonstration and individual defence

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- Expected artifacts: `results/`

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- Saved fallback trace: results/finalnb.pdf.